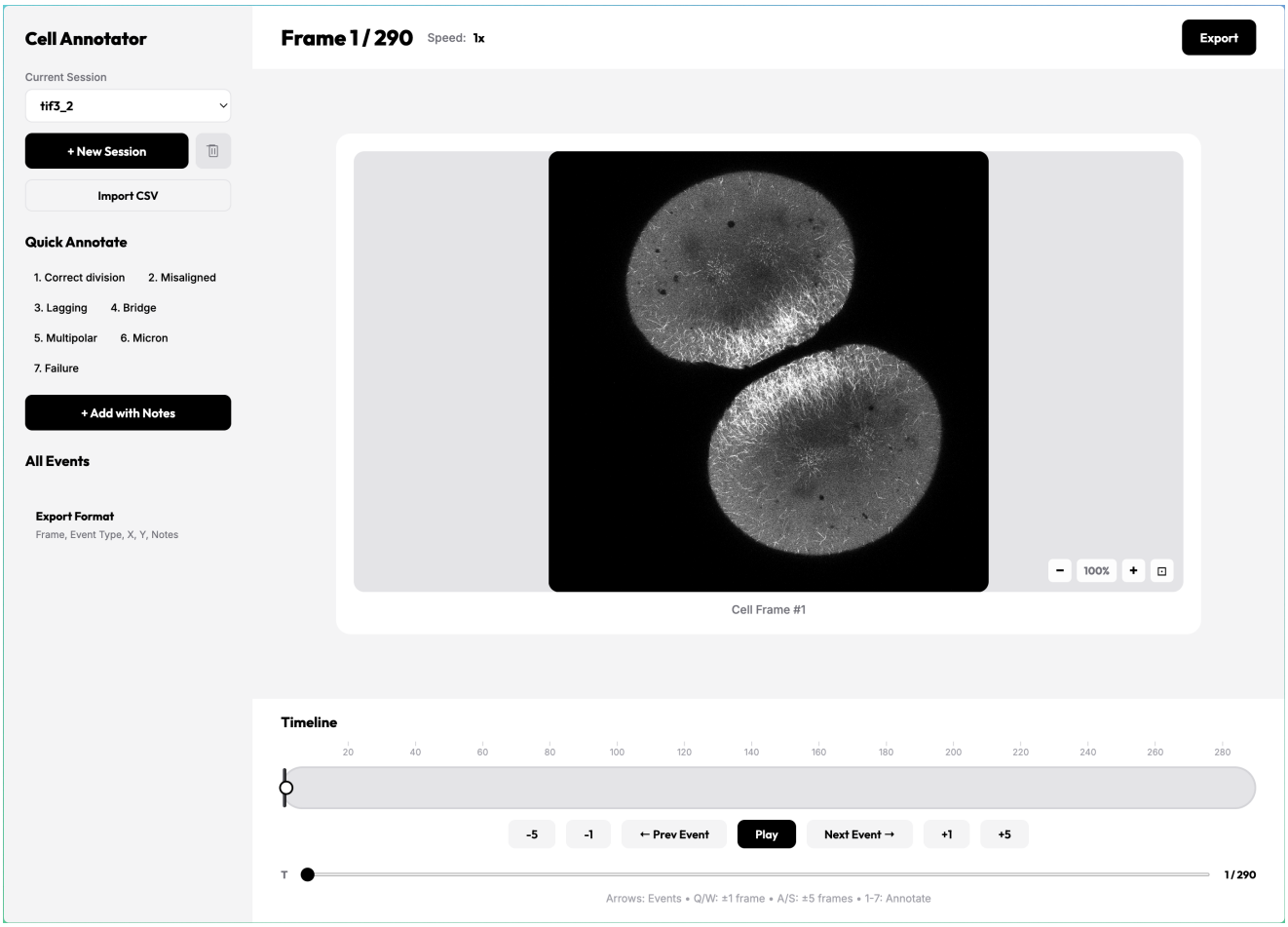
Aneuploidy Annotator

User Guide



Aneuploidy Annotator — main interface

Overview

Aneuploidy Annotator is a browser-based tool for annotating mitotic events in microscopy videos and multi-dimensional TIFF files (including ImageJ hyperstacks and OME-TIFF). Annotations are auto-saved in the browser and can be exported as a ZIP containing a CSV, metadata, and frame images.

1 · Loading a File

- Open **aneuploidy-annotator.html** in a modern browser (Chrome recommended).
- Click **+ New Session**, enter a name, and optionally upload a file.
- Supported formats: **MP4 / video**, **multi-frame TIFF**, **OME-TIFF**, **ImageJ hyperstack**, PNG, JPEG.
- To load a file later, click the image placeholder area to open the file picker.
- Sessions and video files are persisted in browser storage — they survive page reloads.

2 · Navigation

Action	Key / Control
Skip ±1 frame	Q / W
Skip ±5 frames	A / S
Jump between events	← / → arrows

Play / Pause	Space
Scrub timeline	Click or drag timeline bar
Quick annotate type 1–7	1 – 7 number keys
Open Add Event panel	N
Edit event at current frame	E
Delete event at frame	Delete / Backspace
Close panel / cancel	Escape

3 · Z / T / C Sliders (TIFF hyperstacks only)

When a multi-dimensional TIFF is loaded, sliders appear below the timeline:

- **T** — time frame (also controlled by the main timeline scrubber)
- **Z** — Z-slice (optical depth)
- **C** — channel

The header bar shows the current position, e.g. *Frame 5 / 100 | Z 3/10 C 1/2*.

4 · Annotating Events

- Press **1–7** to select an event type, then click on the image to place a marker at exact coordinates.
- Click **+ Add with Notes** to open the event panel for type, notes, and coordinate picking.
- In the event panel, click **"Click to mark on image"** — the cursor changes to a crosshair. Click anywhere on the image to record coordinates. The panel stays open so the image remains visible.
- Click an event card in the sidebar to jump to that frame.
- Hover an event card to reveal ✎ **Edit** and **× Delete** buttons.

Event Types

Key	Event Type
1	Correct division
2	Misaligned chromosomes
3	Lagging chromosomes
4	Anaphase bridge
5	Multipolar spindles
6	Micronuclei
7	Mitosis failure

5 · Zoom & Pan

- Scroll the mouse wheel over the image to zoom in/out (centered on cursor).
- Click and drag to pan when zoomed in (cursor becomes a hand).
- Use the – **+** overlay buttons (bottom-right of image) or ■ to reset zoom.

6 · Exporting

- Check the **checkbox** on any event card to mark that frame for image export.
- Click **Export** (top right) to download a **.zip** file containing:
  - **session.csv** — all annotations (Frame, Type, X, Y, Notes, ExportFrame)
  - **session_meta.txt** — session info and statistics

- **frames/** — PNG images for checked events, named *session_frame42_correct-division.png*

7 · Session Management

- Use the dropdown at the top of the sidebar to switch between sessions.
- Click the **trash icon** next to the dropdown to delete the current session.
- Click **Import CSV** to import annotations from a previously exported CSV.
- All changes are auto-saved — no manual save required.